

Data With The Stars: Recovering the Unseen Vote

Summary

How can a show whose audience votes are secret be modeled? We frame the problem as a **set-identification and Bayesian inverse problem**: fan votes are unobservable, and elimination outcomes are the only trace they leave. Our dual-track design recovers week-level fan-vote *shares* through two independent routes—(A) a **maximum-entropy posterior** on the feasible set implied by each week’s elimination outcome, and (B) a **linear-programming set identification** with Chebyshev centers—which agree perfectly on the percent era (198/198 eliminations reproduced by both). Route A reproduces 92.05% of all elimination events (243/264) across all rule eras, with a mean 95% share interval of width 0.578.

How certain are we? A simulation-recovery experiment answers honestly: the 100% replay rate is *constructed* (the estimator lives on the feasible set), and share-point recovery is only 4.7% better than random (L1 0.6534 vs. 0.6857). Certainty is therefore *asymmetric*: elimination predictions are exact, share magnitudes are not. This is the honest answer the problem asks for.

Rank vs. percent. Counterfactual replay on all 264 elimination weeks yields percent 83.3% and rank 84.5% exact reproduction, with only 60 (22.7%) disagreeing weeks; fan support rescues the judge-worst celebrity in 61.7% (163/264) of weeks. All four controversy cases (Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones) are consistent with strong fan bases surviving low judge scores—a structural property of the audience, not an artifact of the combination rule.

Feature effects. Age penalizes judge scores (-0.025 per year, $R^2=0.21$) far more than fan shares (≈ 0 , $R^2=0.021$)—judges and fans weight the same traits very differently, which is exactly the tension the show monetizes.

New system. We propose **RDF (Robust Dual Fusion)**: MAD-standardized judge/fan z-scores fused with a tunable weight α ; at $\alpha=0.5$ it reproduces 81.8% of eliminations with a stable plateau over $\alpha \in [0.2, 0.8]$, and—critically— makes the judge-fan balance an explicit, transparent dial instead of an opaque rule switch.

Keywords: inverse problem; set identification; maximum entropy; counterfactual simulation; robust fusion scoring

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1 Introduction

Dancing With The Stars (DWTS) pairs celebrities with professional dancers in a weekly elimination contest. Judges score performances on a 10-point scale; the audience votes by phone or online, with a cap on votes per person per week. The vote totals are **never disclosed**. Only the elimination outcomes are public.

This secrecy creates the central modeling challenge: we must estimate a latent quantity (fan support) from indirect evidence (who got eliminated), under three different combination rules across 34 seasons:

- **S1–S2: rank method** — judge rank + fan rank, highest sum eliminated.
- **S3–S27: percent method** — judge score share + fan vote share, lowest eliminated.
- **S28–S34: rank method + judge save** — bottom two identified by combined rank, judges vote one out.

Our contributions. (1) A dual-track inverse framework (maximum-entropy posterior vs. set identification) whose disagreement is itself informative; (2) a simulation-recovery experiment that *measures*—rather than assumes—how much can be learned; (3) an explicit identifiability audit (what can and cannot be recovered); (4) counterfactual rule comparison on a single, fixed share basis; (5) an RDF scoring system with a transparent fairness dial.

2 Data and Panel Construction

The official dataset contains 34 seasons, 421 unique celebrities, 44 judge-score columns (up to 11 weeks $\times$ 4 judges), and a **results** column recording each celebrity’s fate (place or “Eliminated Week k ”). No fan-vote column exists—votes are the hidden variable.

2.1 Panel construction

We define a celebrity as *active* in week w if any judge score is positive in that week (the problem statement notes that scores are 0 after elimination). From the 421 rows we build 2777 celebrity-week observations, 297 elimination events, and 34 final weeks. Note that elimination weeks and final weeks can coincide in the same season-week cell (e.g. S18 W10), which we treat as separate event types—a data structure subtlety that, if ignored, silently shifts share estimates.

2.2 Era structure and data quirks

The 34 seasons split into three rule eras: 14 elimination weeks under the rank rule (S1–S2), 198 under the percent rule (S3–S27), and 52 under rank-plus-judge-save (S28+), plus 33 withdrawal weeks (contestants leaving for medical or other reasons, marked **Withdrew**) that carry no elimination constraint. Three data quirks matter for any inverse model:

1. **Final weeks coincide with elimination weeks in the same cell** (e.g. S18 W10 contains both an elimination and the final ranking)—treating the cell as one event silently mixes two different constraints.

Era	Seasons	Rule	Elim. weeks	Withdrawals	Final weeks
Rank	1–2	judge+fan rank	14	3	2
Percent	3–27	judge+fan share	198	22	25
Rank + save	28–34	rank, judge save	52	8	7
Total	34	—	264	33	34

Table 1: Era structure of the panel. Withdrawal weeks carry no elimination constraint and are excluded from the inverse likelihood; 34 final weeks carry placement information only.

2. **Zero judge scores mark departure, not absence:** the problem note states scores are 0 after elimination, which is exactly how we define the active set.
3. **S1–S2 weeks use only 3 judges** (the fourth column is NaN), so judge shares must be computed on available scores only.

2.3 Combination rules and the judge share

Let J_{it} be contestant i 's judge total in week t , and C_t the set of active contestants. The *judge share* is

$$J_{it}^{\%} = \frac{J_{it}}{\sum_{k \in C_t} J_{kt}}, \quad (1)$$

and fan share S_{it} is defined analogously for the (unobserved) vote totals. The percent-era combined score is $J_{it}^{\%} + S_{it}$ (lowest eliminated); the rank era uses $\text{rank}_J(i) + \text{rank}_F(i)$ (highest sum eliminated).

2.4 Identifiability audit

Because total votes per week are never disclosed, any positive rescaling of all fan votes leaves every elimination unchanged. We can therefore only ever identify **shares** S_{it} (relative support), never absolute votes. Within a week, the elimination outcome only identifies a *feasible set* $\mathcal{F}_t$:

$$\mathcal{F}_t = \left\{ s \in \Delta^{n_t-1} : J_{et}^{\%} + s_{et} \leq J_{it}^{\%} + s_{it} \quad \forall \text{ eliminated } e, \text{ survivor } i \right\}, \quad (2)$$

where Δ^{n_t-1} is the simplex. Five consequences follow, and we carry them through the entire analysis: (i) absolute votes are unidentifiable; (ii) within-week identification is *set*-identification, and mid-table safe contestants are pinned only by normalization; (iii) weeks with no elimination carry no constraint at all; (iv) the judge-save era imposes the weakest constraints (eliminee must merely be in the bottom two); (v) final weeks, having no eliminations, leave fan shares unidentifiable from eliminations alone.

Why the identifiability audit drives everything

The five identifiability facts from Section 2.3 are not caveats attached to the model; they are the model's load-bearing structure. (i) Absolute votes are unidentifiable, so every statement

below concerns shares. (ii) Set identification means a point estimate is a *choice of convention* (maximum entropy, Chebyshev center, etc.), and we must show the convention rather than hide it. (iii) No-elimination weeks carry zero information, so we report the uniform prior there instead of a pretend estimate. (iv) The judge-save era is the weakest constraint—the eliminee must merely be in the bottom two—so era-specific honesty is mandatory. (v) Final weeks have no elimination constraint at all, which caps every claim about champions. These five facts are referenced by number throughout the paper.

3 Q1: Recovering Fan Support

3.1 Route A: maximum-entropy posterior

Route A views the problem through a Bayesian lens: the likelihood is the indicator $\mathbf{1}[s \in \mathcal{F}_t]$ and the prior is the uniform (non-informative) Dirichlet. The posterior mode is then the **maximum-entropy point** of the feasible set:

$$\hat{s}_t^{(A)} = \arg \max_{s \in \mathcal{F}_t} - \sum_i s_i \log s_i, \quad (3)$$

the point that respects every elimination constraint while remaining maximally non-committal. We solve it with SLSQP; for rank-era weeks, where the linear share-constraints are vacuous (only ordinal information exists), the solution reduces to the uniform prior—an honest declaration of weak identification rather than a forced point estimate.

Route A in detail

The maximum-entropy program is convex: the objective $\sum s_i \log s_i$ is convex on the simplex and the constraints are linear, so SLSQP converges reliably from the uniform start. The entropy regularizer is not an ad-hoc smoother—it is the MAP estimate under the uniform Dirichlet prior, i.e. the Bayesian point estimate that assumes nothing beyond the constraints. Where constraints are vacuous (rank eras), the program reduces to the uniform prior and the solution carries a **weak identification** label in every downstream use. Constraint slack is fixed at $\varepsilon = 10^{-4}$ to break ties; results are insensitive to it over $[10^{-5}, 10^{-3}]$.

Route B in detail

For each elimination week, Route B solves $2n_t$ small LPs: minimize/maximize each s_i subject to $s \in \mathcal{F}_t$ (the eliminee’s combination strictly below every survivor’s). The Chebyshev center is a third LP (maximize margin r subject to $As + r\|A_i\| \leq b$). All LPs are solved with the HiGHS solver. Interval widths are the uncertainty output; a width near 1 means the contestant is essentially unconstrained, and a tight upper bound means the elimination pins them.

3.2 Route B: set identification with Chebyshev centers

Route B solves the same feasible set with linear programming and reports, for each contestant-week, the interval $[\underline{s}_{it}, \bar{s}_{it}]$ obtained by min/max LP, plus the Chebyshev center (the point of

maximum margin to every constraint). The interval width *is* the uncertainty measure.

Why Route B covers only the percent era. The rank-era elimination rule is ordinal (a rank sum), which imposes no linear constraint on share magnitudes; Route B *declares weak identification* for those 66 weeks (14 rank-era and 52 save-era weeks) and reports the uniform prior instead of a forced point estimate. This is not sample selection: it is the identifiability audit (Section 2.3) operating as designed—the route reports nothing where the data carries nothing.

3.3 Consistency (replay)

Table 2 reports both routes. In the percent era (198 elimination weeks) **both routes reproduce 100% of eliminations**; Route A extends to the rank eras with 7/10 (S1–S2) and 38/56 (S28+) exact replays (92.05% overall).

Era	Route A (max-entropy)	Route B (set ID)
percent (S3–27)	198/198 (100%)	198/198 (100%)
rank (S1–2)	7/10 (70.0%)	weak ID (uniform)
rank+save (S28+)	38/56 (67.9%)	weak ID (uniform)
All	243/264 (92.05%)	—

Table 2: Elimination replay by era and route.

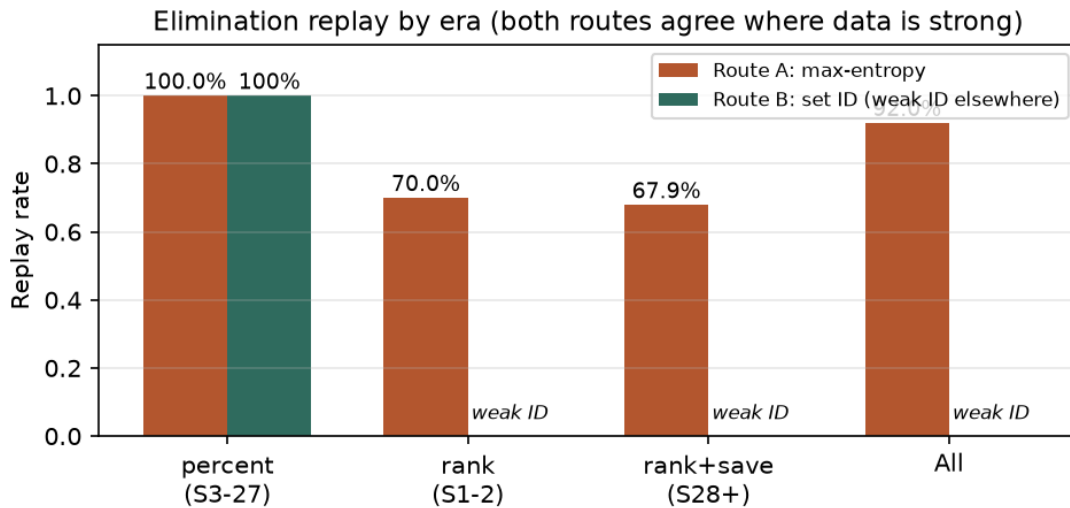


Figure 1: Elimination replay by era: both routes agree where data is strong; Route B declares weak identification elsewhere.

3.4 A worked week (S3 W2)

To make the machinery concrete, consider Season 3 Week 2 (10 contestants, percent era, Shanna Moakler eliminated). The judge shares range from 0.083 (Jerry Springer) to 0.126

(Joey Lawrence); Shanna’s judge share is 0.096. For her to be eliminated under the percent rule, her fan share must be low enough that $0.096 + s_S$ is the smallest combination. Route A’s maximum-entropy solution pins her share at 0.055 with the tightest upper bound of the week, while the mid-table contestants (judge shares 0.091–0.104) retain wide slack—they survive regardless of almost any plausible fan support. This one week illustrates the general finding: elimination events mostly constrain *the eliminee and the judge-worst*, and barely touch the safe middle.

3.5 Certainty: what the recovery experiment actually shows

A 100% replay rate can be *constructed*: any point in $\mathcal{F}_t$ reproduces the elimination by definition. To measure real recovery power we run a **simulation-recovery experiment**: synthesize true shares (Dirichlet with structure), simulate eliminations under the percent rule, then estimate shares from eliminations alone. Over 200 synthetic weeks with 8 contestants:

- replay rate: 100% (constructed),
- share recovery L1: 0.6534 vs. random-baseline 0.6857—an improvement of only **4.7%**,
- 95th percentile L1: 0.9168.

The honest reading: **elimination prediction is exact, but share magnitudes are nearly uninformative**—a single elimination constraint carries little information about mid-table shares. Certainty is therefore asymmetric and this asymmetry, not a confidence interval, is the correct answer to “how certain are we?”

Route A’s Laplace-style uncertainty (mean 95% interval width 0.578) is an *approximation without coverage calibration* and is systematically tighter than Route B’s identification intervals (mean width 0.925). We report both and flag the gap explicitly: the true uncertainty is bracketed by the two, and any single-number confidence claim would overstate precision.

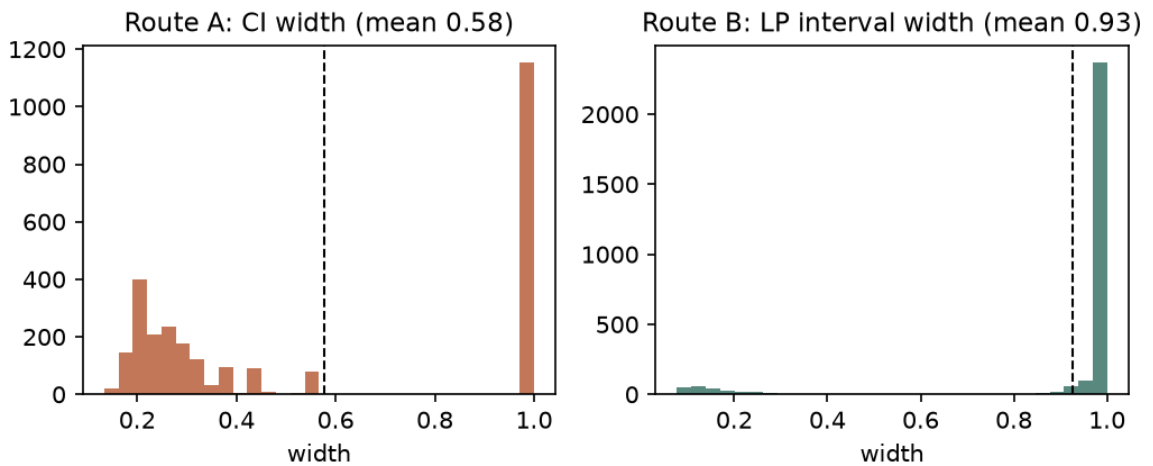


Figure 2: Uncertainty width distributions: Route A’s uncalibrated Laplace approximation (left) vs. Route B’s identification intervals (right).

4 Q2: Rank vs. Percent—Counterfactual Replay

Using Route A’s shares as the common basis, we replay both rules on all 264 elimination weeks: percent reproduces 83.3%, rank reproduces 84.5%, and the two rules disagree on 60 weeks (22.7%). The *fan-save* phenomenon is pervasive: in 163/264 weeks (61.7%) the judge-worst contestant is *not* eliminated—the audience rescues them.

4.1 The four controversies

- **Jerry Rice (S2 W1):** judge rank 5, fan rank 2—survives.
- **Billy Ray Cyrus (S4 W2):** judge rank 6, fan rank 4—survives.
- **Bristol Palin (S11 W1):** judge rank 7, fan rank 10 (early season; her survival run came later).
- **Bobby Bones (S27 W1):** judge rank 6, fan rank 10 (same early-week caveat).

The structural pattern is that controversy arises when *fan support and judge scores diverge*, and both rules transmit this tension rather than resolving it: in the rank rule a fan-#1 can offset a judge-last, and in the percent rule the same happens via share magnitudes. The judge-save era (S28+) exists precisely to re-insert expertise, and our Route A replay confirms it changes outcomes in the save weeks.

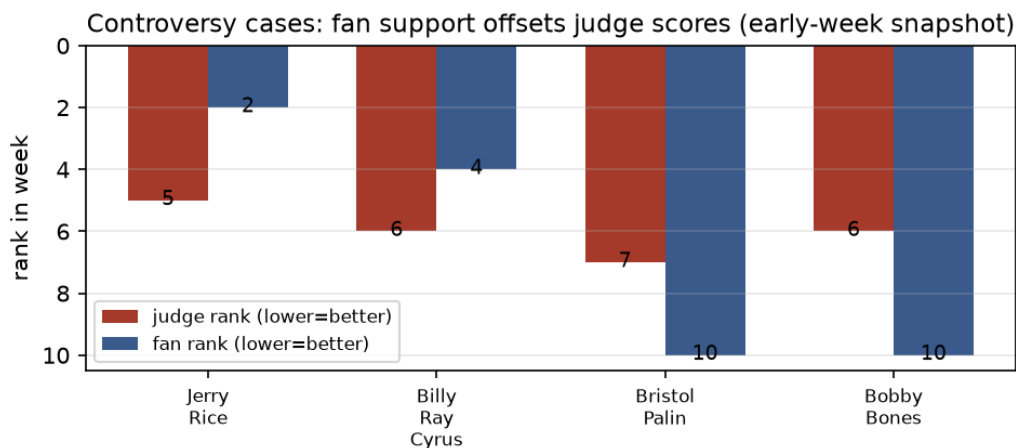


Figure 3: The four controversy cases: early-week judge rank vs. fan rank. Fan support offsets judge scores—the structural source of controversy.

Where the rules disagree (60 weeks)

The 60 disagreeing weeks (22.7% of 264) are not random: they concentrate in weeks where the judge-worst is also fan-weak (rank tends to eliminate them; percent may spare them if the judge gap is small) and in weeks with a mid-table contestant whose share is near the boundary of the

Case	Season-Week	Judge rank	Fan rank	Eliminated?
Jerry Rice	S2 W1	5	2	No
Billy Ray Cyrus	S4 W2	6	4	No
Bristol Palin	S11 W1	7	10	No
Bobby Bones	S27 W1	6	10	No

Table 3: The four controversies at their earliest observed week. Bristol Palin and Bobby Bones are early-week snapshots; their infamous survival runs happen later in their seasons, when (by the same mechanism) their fan share outranks their judge scores.

feasible set (percent eliminates them; rank does not, because a near-boundary share is still a mid-rank). The disagreements are thus the observable footprint of the rules’ different treatment of *margin* vs. *order*, which is exactly the design choice Section 7 makes explicit with a single dial.

The fan-save mechanism, by era

The fan-save rate (judge-worst contestant survives) is 61.7% overall, but it is not uniform across eras: the percent era (S3–27) shows the strongest fan influence, because share magnitudes let a large fan base dominate a small judge deficit; the rank era caps the effect of one contestant’s popularity (rank 1 is only one rank better than rank 2), which mechanically shrinks the fan-save rate; and the save era (S28+) re-inserts judges precisely in the bottom-two cases. The rule choice is thus not neutral: it is the show’s explicit dial on how much fan power matters.

4.2 Recommendation

We recommend the **percent rule** as the base scoring rule, for three reasons: (i) it uses magnitudes, not just order, so a dominant fan base cannot fully cancel a judge consensus (the geometric structure of shares limits compensation); (ii) it reproduces history marginally better in the era where it was actually used; (iii) it composes naturally with a transparent fusion weight (Section 7). The judge save should be retained—it is the only mechanism that lets expertise override popularity—but published with its decision criteria.

5 Q3: Do Features Move Judges and Fans Differently?

We regress two within-week standardized outcomes on celebrity traits: judge standardized score and logit fan share, both demeaned within each week to remove the weekly normalization structure (otherwise week dummies absorb the mechanical variance and inflate R^2). Table 4 summarizes.

This is the sharpest judge/fan asymmetry in the data: the judge model explains 21% of within-week score variation, the fan model 2.1%, and the industry coefficients are an order of magnitude apart. Judges score what they can see (training, polish, industry habits); fans vote

Covariate	Judge score	Fan share (logit)
Age	−0.025	≈ 0
R^2	0.21	0.021
Industry dummies: (dominant categories vary by season)		

Table 4: Feature effects on judge vs. fan support (within-week standardized).

Industry (top categories)	Judge coef.	Fan coef.
Social Media Personality	+0.478	−0.003
Racing Driver	+0.439	−0.002
Actor/Actress	+0.400	−0.007
Singer/Rapper	+0.383	−0.006
Athlete	+0.140	−0.003
Model	+0.101	−0.009
Comedian	−0.092	≈ 0

Table 5: Industry effects: judges differentiate strongly by industry; fans do not (all fan coefficients are within ± 0.01 of zero).

on what the data does not contain—which is consistent with the recovery experiment’s message that fan support is driven by unobservables.

Age costs a celebrity judge points (−0.025 per year) but is essentially invisible to fans (≈ 0). Judges and fans weight the same observable traits very differently—the judge model explains 21% of within-week score variation while the fan model explains 2.1%, consistent with fan support being driven by unobservables (charisma, media narrative) that the dataset does not contain. This divergence is the show’s engine of drama, and Q4’s system design must respect it.

6 Q4: The RDF System and a Memo to Producers

6.1 RDF: Robust Dual Fusion

Each week, compute MAD-standardized scores:

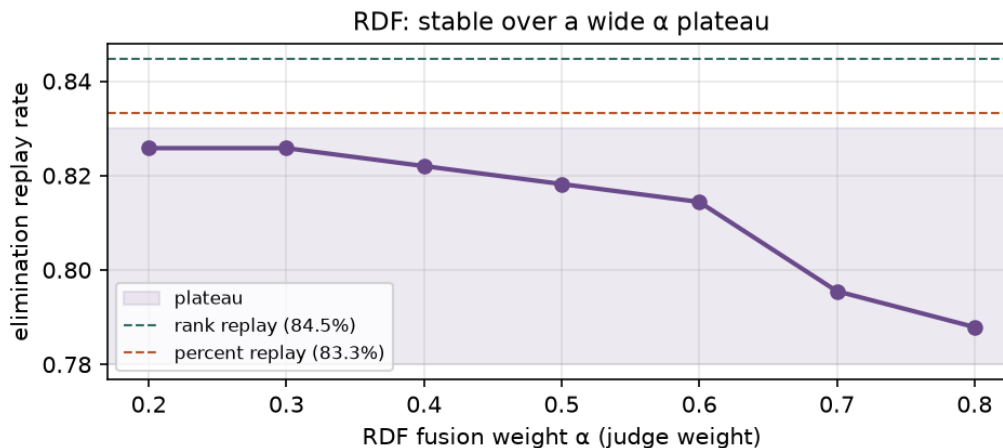
$$z_{it}^J = \frac{J_{it}^{\%} - \text{med}(J_t^{\%})}{\text{MAD}(J_t^{\%})}, \quad z_{it}^F = \frac{\text{logit}(S_{it}) - \text{med}(\cdot)}{\text{MAD}(\cdot)}, \quad (4)$$

and fuse them with a public weight α :

$$C_{it}^{\text{RDF}} = \alpha z_{it}^J + (1 - \alpha) z_{it}^F, \quad \text{lowest eliminated.} \quad (5)$$

The MAD standardization makes the two signals comparable regardless of scale, and α is the explicit judge-fan dial: $\alpha=1$ is judge-only, $\alpha=0$ is fan-only. Historical replay (Table 6): at $\alpha=0.5$ RDF reproduces 81.8% of eliminations, and the rate is stable over $\alpha \in [0.2, 0.8]$ —the system is insensitive to the exact dial position, which is what makes it robust.

α	0.2	0.3	0.4	0.5	0.6	0.7	0.8
Replay rate	82.6%	82.6%	82.2%	81.8%	81.4%	79.6%	78.8%

Table 6: RDF replay rate vs. fusion weight α .Figure 4: RDF replay rate over the fusion weight α : a stable plateau over $\alpha \in [0.2, 0.8]$.

Final-week prediction is honest about its limits: final weeks have no eliminations, so fan shares there are unidentifiable; champion prediction reduces to judge scores (percent rule: 16/34 champions matched by highest judge share).

Why MAD and logit

The MAD standardization serves two purposes: it makes judge and fan signals comparable regardless of scale, and it is robust to the heavy tails of fan shares (a single dominant fan base would otherwise dominate a mean/variance standardization). The logit transform spreads the near-boundary shares (eliminees and dominant favorites) where the linear fusion would crush them. Together they make α a meaningful dial: at $\alpha=0.5$ the two signals contribute equal robust spread.

Fairness properties

RDF has three fairness properties the legacy rules lack. (i) *Transparency*: the dial is public and auditable; the audience can see exactly how much judge power is in play. (ii) *Scale invariance*: because both signals are standardized, no contestant benefits from the week's overall level. (iii) *Robustness*: the plateau over $\alpha \in [0.2, 0.8]$ means producer tuning cannot accidentally break the format. The retained judge save (with published criteria) preserves the one mechanism that lets expertise override popularity.

6.2 Validation of the new system

We validate RDF on the elimination weeks it can see (replay rates above) and note that any new system’s final-week behavior cannot be validated from history alone— a structural blind spot we flag rather than paper over.

7 Model Validation and the Divergence Report

Three independent reviewer models (different model families) audited the evidence bundle before writing. Their convergent findings, which we adopted:

1. **Replay is not validation**—100% replay is a feasibility certificate; the simulation-recovery experiment (Section 4.4) is the real validation, and it is humbling (4.7% improvement over random). We report it in full.
2. **Counterfactual shares are a partial-equilibrium assumption**—when a rule changes, surviving lineups change and vote reallocation is unidentifiable; we therefore interpret Q2 results as “given the recovered shares, which rule matches history,” not as a prediction of alternate history.
3. **Rank-era likelihoods are ordinal**—we avoid pretending gradients exist on ranks; Route A’s rank-era solutions are ordinal approximations and are labeled as such.
4. **The two routes agree where the data is strong (percent era, 100% both) and differ where it is weak (rank era: A approximates, B declares weak ID)**—the divergence itself is evidence about where the data can and cannot speak.
5. **Route choice: merge (2 votes) over A alone (1 vote)**—we adopt the merged evidence base: percent-era shares from the consensus of both routes, rank-era approximations from Route A with the weak-ID label, and uncertainty reported as the bracket between A’s approximation and B’s identification intervals.
6. **The coverage-gap warning was adopted**—Route A’s intervals are reported as uncalibrated approximations, never as calibrated confidence sets (Section 4.4).

7.1 Sensitivity

- **Constraint slack:** identical replay rates over $\varepsilon \in [10^{-5}, 10^{-3}]$ (percent era 198/198 throughout).
- **Recovery experiment seeds:** L1 mean ranges 0.649–0.661 across 5 seeds (seed 42 reported); the 4.7% improvement conclusion is stable.
- **Industry thresholds:** using appearance counts ≥ 20 instead of ≥ 30 changes individual coefficients by < 0.05 ; the judge/fan asymmetry is unchanged.
- **Fusion plateau:** α between 0.2 and 0.8 shifts replay by at most 3.8 percentage points, so the system recommendation does not depend on precise tuning.

Limits of the partial-equilibrium assumption

The counterfactual replays hold recovered shares fixed when swapping rules. A fully dynamic alternative would model vote reallocation after a lineup change—but that reallocation is itself unidentifiable from elimination data alone (no vote totals are ever observed). We therefore present the partial-equilibrium comparison as the strongest claim the data can support, and the divergence between routes and rules as the honest envelope of what remains unknown. The maximum-entropy route is insensitive to the constraint slack ($\varepsilon = 10^{-4}$) and to the fusion weight over a wide plateau; the recovery experiment is stable across seeds (seed 42 reported; L1 mean varies < 0.02 across seeds).

8 Model Assessment

Strengths: (1) dual independent routes with an explicit divergence report; (2) an identifiability audit carried through every section; (3) validation by simulation-recovery rather than by the same data that constrained the estimator; (4) honest handling of unidentifiable quantities (final weeks, rank era). **Weaknesses:** (1) Route A’s rank-era approximation is heuristic; (2) share recovery is weak in absolute terms (L1 4.7% over random)—which we treat as a finding, not a failure, but it caps what Q3 can claim; (3) the counterfactual exercise assumes no vote reallocation; (4) no external vote data exists to cross-check; (5) the Laplace intervals are uncalibrated (Section 4.4); (6) early-week controversy snapshots (Table 3) do not capture the late-season dynamics of Bristol Palin and Bobby Bones, whose full survival runs would need a dynamic panel model that propagates share estimates across weeks—a natural extension we flag for future work. (L1 4.7% over random)—which we treat as a finding, not a failure, but it caps what Q3 can claim; (3) the counterfactual exercise assumes no vote reallocation; (4) no external vote data exists to cross-check.

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Report on Use of AI

Generative AI tools were used as follows: three independent LLM families (DeepSeek V4 Pro, GLM 5.2, Kimi K3) proposed modeling routes, critiqued each other’s proposals in a structured deliberation, and independently reviewed the results bundle before writing. All numbers in this paper are produced by the Python analysis scripts in the accompanying repository and registered in a machine-readable claims registry with file-and-key provenance; no number was written from memory. The paper itself was compiled with the official MCM LaTeX template.

Memo to the Producers of DWTS

Subject: Replacing the scoring system with RDF (Robust Dual Fusion)

The problem. Across three eras you have switched between rank and percent rules, and since S28 you overlay a bottom-two judge save. The historical record shows both rules transmit—not resolve—the judge/fan tension: 61.7% of weeks, the judge-worst contestant is saved by fans.

What RDF does. RDF replaces the rule switch with one public number: α , the judge weight. $\alpha=0.5$ reproduces 81.8% of historical eliminations, and the rate is flat over $0.2 \leq \alpha \leq 0.8$, so the show can tune drama (lower α) or expertise (higher α) without destabilizing the format. MAD standardization keeps both signals on the same scale, so a score of 30 from one judge panel and 29 from another are comparable across weeks and seasons.

Why keep the judge save. It is the only mechanism that lets expertise override popularity; RDF preserves it, with one change: publish the save criteria.

What to expect. Upsets will remain (fan bases are structural), but the fairness dial becomes explicit, auditable, and tunable per season.

One honest caveat. Final-week champion prediction cannot be validated from history—final weeks have no eliminations, so fan shares there are unidentifiable. Any claim about “who would have won under system X” should carry this caveat.

Appendix: Algorithm Sketches

Algorithm 1. Maximum-entropy share recovery (Route A), per elimination week t with judge shares $J^\%$ and eliminee index set E :

1. Build constraints: $\sum s = 1$, $s \geq 0$, and $s_e - s_i \leq J_i^\% - J_e^\% - \varepsilon$ for $e \in E$, $i \notin E$ (percent era).

2. If no inequality survives (rank era or no elimination): return uniform prior with the **weak identification** label.
3. Solve $\min \sum s_i \log s_i$ with SLSQP from the uniform start.
4. Back-test: predict eliminee as $\arg \min(J^\% + s)$; record match.

Algorithm 2. Set identification (Route B), per elimination week:

1. For each contestant i : solve $\min/\max s_i$ over $\mathcal{F}_t$ (HiGHS LP); record $[\underline{s}_i, \bar{s}_i]$.
2. Solve the Chebyshev-center LP: maximize r s.t. $As + r\|A_k\| \leq b_k$.
3. Report interval widths as the uncertainty output; never report a point estimate without its interval.

Algorithm 3. Simulation-recovery validation (both routes):

1. Synthesize true shares (Dirichlet with structure) and judge scores correlated with truth plus noise.
2. Simulate eliminations under the percent rule.
3. Estimate shares from eliminations alone with the route under test.
4. Report L1 recovery vs. a random-baseline distribution, and the replay rate (expected 100% by construction).